A counterexample to degree-zero universality of the interleaving distance

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Abstract

The universality conjecture for the interleaving distance asks whether every distance stable under uniform perturbations of sublevel functions is bounded by the interleaving distance on realizable persistence modules. Its formulation for each homology degree is false for ordinary zero-dimensional homology. For every field we construct a bounded, 0-stable pseudometric on all four-parameter persistence modules and a pair realized on three-point discrete spaces for which this distance is 3 and the interleaving distance is 1. The algebraic pair is Bjerkevik's published free-module example. The additional argument recovers the upper supports of eventual sublevel components from the image of a module in its colimit and proves stability for arbitrary spaces and functions, with no finite-dimensionality assumption.

1 The degree-zero question

Sublevel persistent homology records how the topology of a space changes as one or several thresholds increase. A distance between the resulting modules is stable when a uniform change in the threshold function bounds the change in the descriptor. Universality asks whether the interleaving distance is the largest distance with this property on modules that arise from homology. Lesnick proved this assertion over prime fields in positive homology degrees [1, Theorem 5.5]. His Conjecture 5.7 includes degree zero and arbitrary fields, with the comparison restricted to realizable modules. We give a counterexample to its ordinary degree-zero clause.

Fix a field k. All modules in this note are functors $M : (\mathbb{R}^4, \leq) \to \mathrm{Vect}_k$, with coordinate-wise order. Write $\mathbf{1} = (1, 1, 1, 1)$ and $M(\varepsilon)_a = M_{a+\varepsilon\mathbf{1}}$. An ε-interleaving, $\varepsilon \geq 0$, consists of natural maps $\varphi : M \to N(\varepsilon)$ and $\psi : N \to M(\varepsilon)$ satisfying

$$\psi(\varepsilon)\varphi = \eta_M^{2\varepsilon}, \qquad \varphi(\varepsilon)\psi = \eta_N^{2\varepsilon},$$

where $\eta_M^{2\varepsilon}$ is the structure map $M_a \to M_{a+2\varepsilon\mathbf{1}}$. The distance d_I is the infimum of such ε, or $+\infty$ if none exists. For any topological space X and arbitrary function $f : X \to \mathbb{R}^4$, let

$$P_0(f)_a = H_0(\{x \in X : f(x) \leq a\}; k),$$

using ordinary singular homology and sublevel inclusions. A pseudometric on module isomorphism classes is 0-stable if

$$d(P_0(f), P_0(g)) \leq \sup_{x \in X} \|f(x) - g(x)\|_\infty \tag{1}$$

for every X, f, g, with the empty supremum equal to zero. Extended pseudometrics may take the value $+\infty$ and may vanish on distinct classes.

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Theorem 1. *For every field k, there is a $[0,3]$-valued pseudometric d_0 on isomorphism classes of all four-parameter k-modules satisfying (1). There are continuous functions f, g on the discrete space $\{1,2,3\}$ such that, for $M = P_0(f)$ and $N = P_0(g)$,*

$$d_I(M,N) = 1, \qquad d_0(M,N) = 3.$$

The finite algebraic separation used here is Bjerkevik's example [2, Examples 5.2–5.3]. A barcode matching distance on free modules alone does not supply the required distance on all module classes. We construct such a distance from the supports of components in the directed colimit. Uniform perturbations match these components, whereas linear interleavings can mix their basis vectors. This distinction gives the inequality in Theorem 1.

The conclusion concerns ordinary H_0. It does not resolve the arbitrary-field question in positive degrees, nor does it contradict prime-field universality for a distance required to be stable in every homology degree [1, Corollary 5.6]. Lesnick's one-parameter pointwise finite-dimensional result in Section 5.4 uses reduced homology; that convention is not used here.

2 Supports in the image of a module's colimit

For any module M, put

$$C_M = \operatorname*{colim}_{a \in \mathbb{R}^4} M_a, \qquad I_M(a) = \operatorname{im}(M_a \longrightarrow C_M).$$

The subspaces $I_M(a)$ increase with a, and their union is C_M. Indeed, the vector-space colimit is the direct sum of the M_a modulo structure-map relations. An element has finitely many representatives, which can be transported to a common upper grade.

Consider modules for which C_M is finite-dimensional and has a basis $e_1, \dots, e_r$ such that every $I_M(a)$ is spanned by a subset of this basis. The zero colimit is included. For each basis vector define its upper support

$$U_j = \{a \in \mathbb{R}^4 : e_j \in I_M(a)\}.$$

An upper set is a subset closed under increasing coordinates. Each U_j is nonempty because the images exhaust C_M. Let $\mathcal{S}(M)$ be the finite multiset of these supports, with one occurrence for each basis vector.

Proposition 2. *The existence of such a coordinate basis and the multiset $\mathcal{S}(M)$ depend only on the module's isomorphism class.*

Proof. We recover each support multiplicity directly from the subspaces $I_M(a)$. For a nonempty upper set U, define

$$K_U = \bigcap_{a \in U} I_M(a), \qquad L_U = \sum_{V \text{ nonempty upper } U \subsetneq V} K_V.$$

The second subspace lies in the first. In a coordinate basis, membership in any intersection can be tested coefficient by coefficient, giving

$$K_U = \operatorname{span}\{e_j : U \subseteq U_j\}, \qquad L_U = \operatorname{span}\{e_j : U \subsetneq U_j\}.$$

For the second identity, a basis vector belonging to K_V has support containing V and therefore properly containing U. Conversely, if $U \subsetneq U_j$, that vector belongs to the summand K_{U_j} of L_U. Thus

$$\dim(K_U/L_U) = \#\{j : U_j = U\}.$$

This recovers the multiplicity of every upper set without choosing a basis. A natural module isomorphism induces a colimit isomorphism carrying every $I_M(a)$ to the corresponding image. It therefore preserves the coordinate basis property and all the recovered multiplicities. □

3 A distance stable for all ordinary degree-zero inputs

For nonempty upper sets U, V, define

$$\rho(U,V) = \inf\{\varepsilon \geq 0 : U + \varepsilon\mathbf{1} \subseteq V, \quad V + \varepsilon\mathbf{1} \subseteq U\}.$$

The infimum of the empty set is $+\infty$. This is an extended pseudometric. To check the triangle inequality, two pairs of shifted containments compose with the shifts added. Taking arbitrarily close upper bounds for the two infima gives the triangle inequality without an attainment assumption.

For finite multisets $\mathcal{U}, \mathcal{V}$ with the same number of occurrences, let

$$\beta(\mathcal{U},\mathcal{V}) = \inf_{\sigma} \max_{U\in\mathcal{U}} \rho(U, \sigma(U)),$$

where σ ranges over occurrence bijections. Set the distance to $+\infty$ for unequal cardinalities and set $\beta(\varnothing,\varnothing) = 0$. Composing bijections and applying the triangle inequality for ρ proves that β is an extended pseudometric. Adjoin a symbol $*$ at distance $+\infty$ from every multiset and at distance zero from itself; write β_* for this extension.

Define a feature on every module class by

$$\mathcal{F}(M) = \begin{cases} \mathcal{S}(M), & C_M \text{ has a finite coordinate basis as above,} \\ *, & \text{otherwise,} \end{cases}$$

Then put

$$d_0(M,N) = \min\{3, \beta_*(\mathcal{F}(M), \mathcal{F}(N))\}.$$

Proposition 2 makes this definition independent of all choices. Pullback preserves the pseudometric properties, and the cap preserves the triangle inequality by $\min\{3, s+t\} \leq \min\{3,s\} + \min\{3,t\}$ for $s,t \in [0,+\infty]$. Consequently d_0 is a bounded pseudometric on the entire class of modules.

Proposition 3. *The distance d_0 is 0-stable for every topological space and arbitrary functions to $\mathbb{R}^4$.*

Proof. Ordinary singular $H_0(Y;k)$ is naturally the free vector space $k[\pi_0(Y)]$ on path components: singular 0-chains are free on points, and boundaries of singular 1-simplices impose precisely the relations identifying endpoints of paths. This description is natural for maps. Write

$$\Pi_a^f = \pi_0(\{x : f(x) \leq a\}), \qquad B_f = \operatorname*{colim}_a \Pi_a^f$$

with the second colimit taken in sets. The free-vector-space construction commutes with this colimit. For any vector space W, compatible linear maps from $k[\Pi_a^f]$ to W are exactly compatible set maps from Π_a^f to the underlying set of W, hence set maps from B_f to W. This universal property gives

$$C_{P_0(f)} \cong k[B_f].$$

The image $I_{P_0(f)}(a)$ is spanned by the subset of B_f reached from Π_a^f. If B_f is finite, it supplies a coordinate basis and its supports are exactly $\mathcal{S}(P_0(f))$. If B_f is infinite, the colimit is infinite-dimensional and $\mathcal{F}(P_0(f)) = *$. We use this actual sublevel colimit throughout; no identification of B_f with $\pi_0(X)$ is needed.

Let $\delta = \sup_x \|f(x) - g(x)\|_\infty$. If $\delta = +\infty$, the required bound is automatic. Otherwise the sublevel inclusions induce natural set maps

$$\Pi_a^f \longrightarrow \Pi_{a+\delta\mathbf{1}}^g, \qquad \Pi_a^g \longrightarrow \Pi_{a+\delta\mathbf{1}}^f.$$

Their composites are the structure maps at shift 2δ. They therefore induce inverse bijections $\alpha : B_f \to B_g$ on the colimits: the composite sends any represented component to its later structure image, representing the same colimit element. Compatibility of the maps makes this independent of the chosen representative.

If these sets are infinite, both features are $*$ and $d_0 = 0$. If finite, α matches their supports with

$$U_b^f + \delta\mathbf{1} \subseteq U_{\alpha(b)}^g, \qquad U_{\alpha(b)}^g + \delta\mathbf{1} \subseteq U_b^f.$$

The first containment follows from the first component map at every grade in U_b^f. The reverse map gives the second, since its colimit bijection is α^{-1}. Thus each matched ρ is at most δ, so $\beta \leq \delta$ and $d_0 \leq \delta$. The argument includes $\delta = 0$. For empty X both multisets are empty and $d_0 = 0$. □

In particular, finite perturbations preserve the finite or infinite cardinality of the component colimit. This is why assigning the exceptional feature to infinite-colimit inputs is compatible with the stability bound. Individual sublevel homology spaces may still be infinite-dimensional when the component colimit is finite.

4 The three-point counterexample

For $p \in \mathbb{R}^4$, let $k_{\uparrow p}$ denote the module with value k at $a \geq p$, zero elsewhere, and identity maps between nonzero values. Take the birth vectors of Bjerkevik's four-parameter free-module example [2, Example 5.3]:

$$\begin{aligned} u_1 &= (0, 1, 10, 11), & v_1 &= (1, 0, 11, 10), \\ u_2 &= (0, -1, 12, 11), & v_2 &= (1, 0, 9, 12), \\ u_3 &= (2, 1, 10, 9), & v_3 &= (-1, 2, 11, 10). \end{aligned}$$

Put $M = \bigoplus_{i=1}^3 k_{\uparrow u_i}$ and $N = \bigoplus_{j=1}^3 k_{\uparrow v_j}$. On the discrete space $X = \{1, 2, 3\}$, define $f(i) = u_i$ and $g(i) = v_i$. Its sublevels consist of the points whose birth vectors lie below the grade, so naturally $P_0(f) \cong M$ and $P_0(g) \cong N$. Both functions are continuous.

A morphism $k_{\uparrow p} \to k_{\uparrow q}(\varepsilon)$ is a scalar map if $p + \varepsilon\mathbf{1} \geq q$, and otherwise it is zero. Indeed, in the first case every source grade lies in the shifted target, and naturality at common upper grades forces one scalar. In the second case the map at p is zero; naturality from p and any other source grade to a common upper grade forces every component to be zero. This also describes each component of a morphism between the finite direct sums.

The least directed shifts for $M_i \to N_j(\varepsilon)$ and $N_j \to M_i(\varepsilon)$ are respectively

$$T^{\rightarrow} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 3 & 1 \end{pmatrix}, \qquad T^{\leftarrow} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

The first matrix has source M_i in row i and target N_j in column j; the second has source N_j in row j and target M_i in column i. Their entries are obtained by taking the maximum coordinate of the target birth vector minus the source birth vector.

The inverse matrices from Bjerkevik's Example 5.2, equation (15), are

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Use the source-row convention: the component of $\varphi : M \to N(1)$ from M_i to $N_j(1)$ has coefficient A_{ij}, and the component of $\psi : N \to M(1)$ from N_j to $M_i(1)$ has coefficient B_{ji}. Every nonzero entry has directed threshold 1, so these are natural transformations. Direct integer multiplication gives

$$AB = \begin{pmatrix} -1+1+1 & 1-1 & 1-1 \\ -1+1 & 1 & 1-1 \\ -1+1 & 1-1 & 1 \end{pmatrix} = I_3, \qquad BA = I_3.$$

Each nonzero composite component follows two allowed support containments. Thus at every active source grade the composite is given by the same constant matrix product, and these products give the structure maps at shift 2. The integer identities hold over every field, including characteristic 2. We have a 1-interleaving.

For $0 \le \varepsilon < 1$, every component map $M_i \to N_j(\varepsilon)$ is zero, as all entries of $T^{\to}$ are at least 1. The map φ must therefore be zero. It cannot compose to $\eta_M^{2\varepsilon}$, which is nonzero at any grade with nonzero M. Consequently $d_I(M, N) = 1$.

The image families of M and N in their three-dimensional colimits are coordinate families. Their support multisets consist of the principal upper sets $\uparrow u_i$ and $\uparrow v_j$. The shifted-containment definition gives

$$\rho(\uparrow p, \uparrow q) = \|p - q\|_\infty.$$

The matching-cost matrix is therefore

$$\left(\|u_i - v_j\|_\infty\right)_{i,j} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 3 \end{pmatrix}.$$

A matching of cost below 3 would have to send both the second and third M occurrences to the first N occurrence. This is impossible for a bijection. Every entry is at most 3, so a matching of cost 3 exists. It follows that $d_0(M, N) = 3$. Together with Proposition 3, this proves Theorem 1: the realizable ordinary-H_0 modules above have stable distance 3 and interleaving distance 1, over every field.

References

[1] M. Lesnick, *The Theory of the Interleaving Distance on Multidimensional Persistence Modules*, Foundations of Computational Mathematics **15** (2015), 613–650. doi:10.1007/s10208-015-9255-y. Author version: arXiv:1106.5305v4.

[2] H. B. Bjerkevik, *Stability of higher-dimensional interval decomposable persistence modules*, arXiv:1609.02086v3, 21 January 2020; Examples 5.2–5.3, pp. 17–18.

AI-use statement

This manuscript was prepared with assistance from OpenAI Codex for source retrieval, proof development, drafting, and checks of the finite integer calculations. The mathematical argument was reviewed by the same AI assistant. Independent human mathematical review is requested.